

Toward Cruelty-Free Antibodies

Thomas Dupic

CIML Week, November 4th 2026

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Antibodies: a successful transition out of animal use?

In the EU, between **3 and 10% of research animals** are used for antibody development and production.

Multiple production methods now coexist: polyclonal vs monoclonal, hybridomas vs recombinant.

Does this lower animal use?

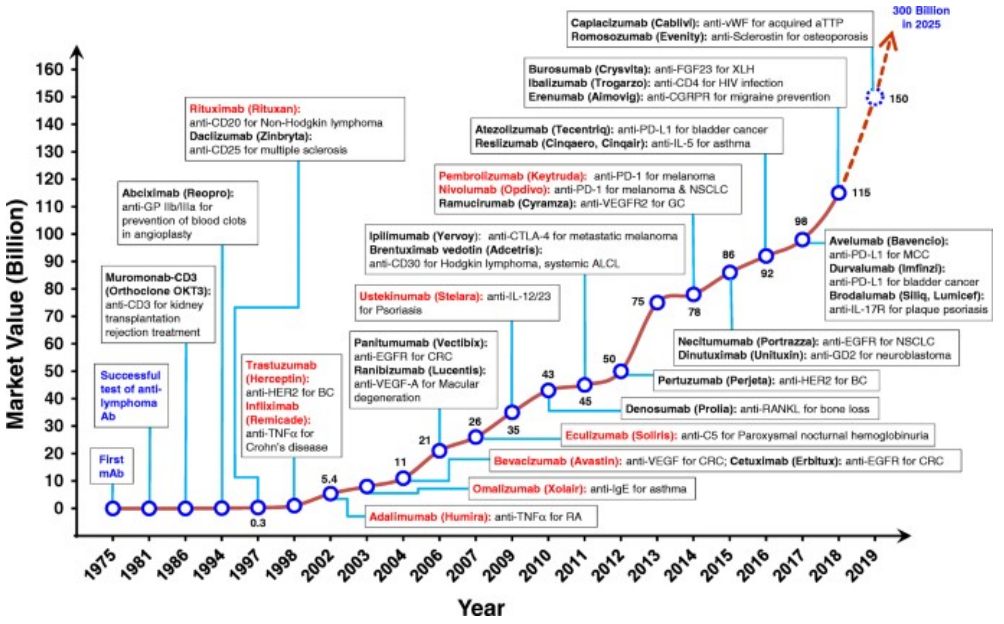


Antibodies as drugs

Antibodies are the largest class of biological drugs by revenue.

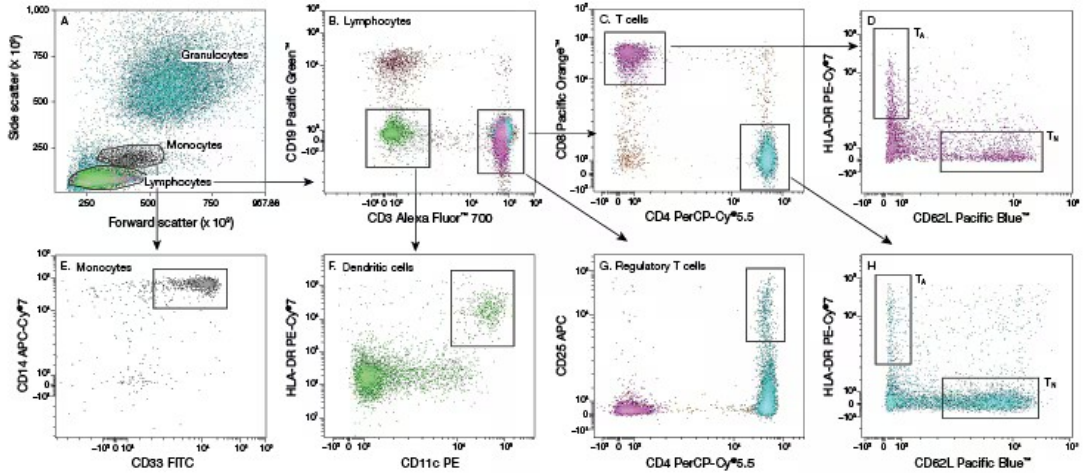
$200 billions in 2024 & many different medical targets.

Adalimumab, Rituximab, Pembrolizumab …



Antibodies as tools

Ubiquitous in biology and immunology: ELISA, Western Blot, Flow Cytometry, Immunoprecipitation, Microscopy …



Also in real life: Pregnancy test, blood typing test, Covid test…

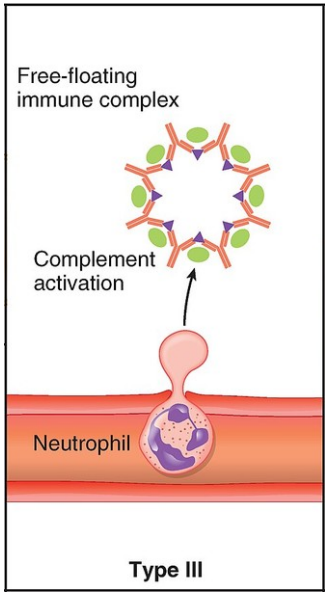
Early days: Serum therapies

Serum Therapies
Behring & Kitasato (1890), diphtheria anti-toxin, produced in horses and guinea pigs



Sera for therapeutics

Inconvenient: Serum sickness (type III hypersensitivity). Therapeutical use survive in antivenoms and some antitoxins.



Human polyclonals: IVIg (mostly plasma)
Primary treatment for immunodeficiency & some auto-immune diseases

Sera for research reagents

Polyclonal antibodies. Mostly rabbits (65%), but also goats, sheep, donkeys, chickens… Primary & secondary.

Immunisation, followed by exsanguination, sera collection and often affinity purification / cross-adsorption.

	Polyclonal	Monoclonal
Initial cost	$	$$$
Tolerance	+	–
Specificity	–	++
Supply & consistency	–	+++

Hybridomas

B cell × Cancer : fusion of B-cells from an immunized animal and myeloma cell line.

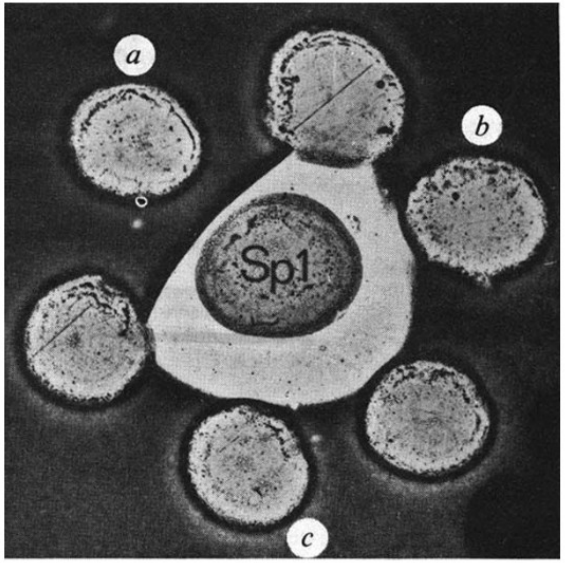
Originally by Sendai virus, now mostly PEG or electro-fusion.

Selection in HAT against the myeloma line (blocks DNA synthesis pathway); B-cells have a limited number of divisions.

HAMA : immune reaction to mouse antibodies. Human × Mice fusions are possible but unstable. “Humanized” mice

Nature Vol. 256 August 7 1975

Continuous cultures of fused cells secreting antibody of predefined specificity



origins[4,5]. Such cells can be grown *in vitro* in massive cultures to provide specific antibody. Such cultures could be valuable for medical and industrial use.

G. Köhler
C. Milstein

Methods of production



Methods of production



Ascites production vs **in-vitro culture**

Injection of hybridoma cells in the inflammed peritoneal cavity of a mouse

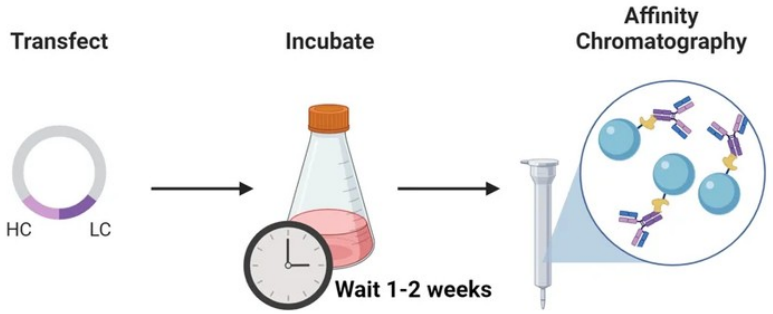
Restricted or prohibited in most European country.

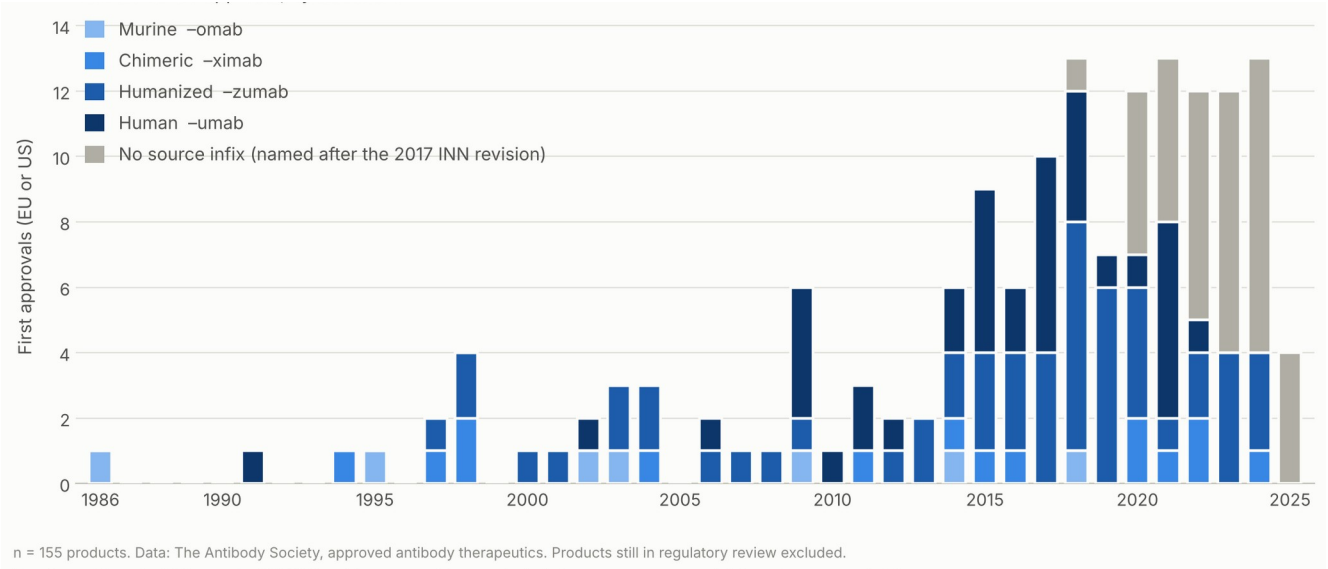
99% of the european production is in France

Standard in-vitro growth (RPMI)

Recombinant antibodies

Expressed from cloned DNA in a transfected cell. Known sequence, no drift, engineerable.





First cloning in 1984, quickly adopted for therapeutic antibodies.

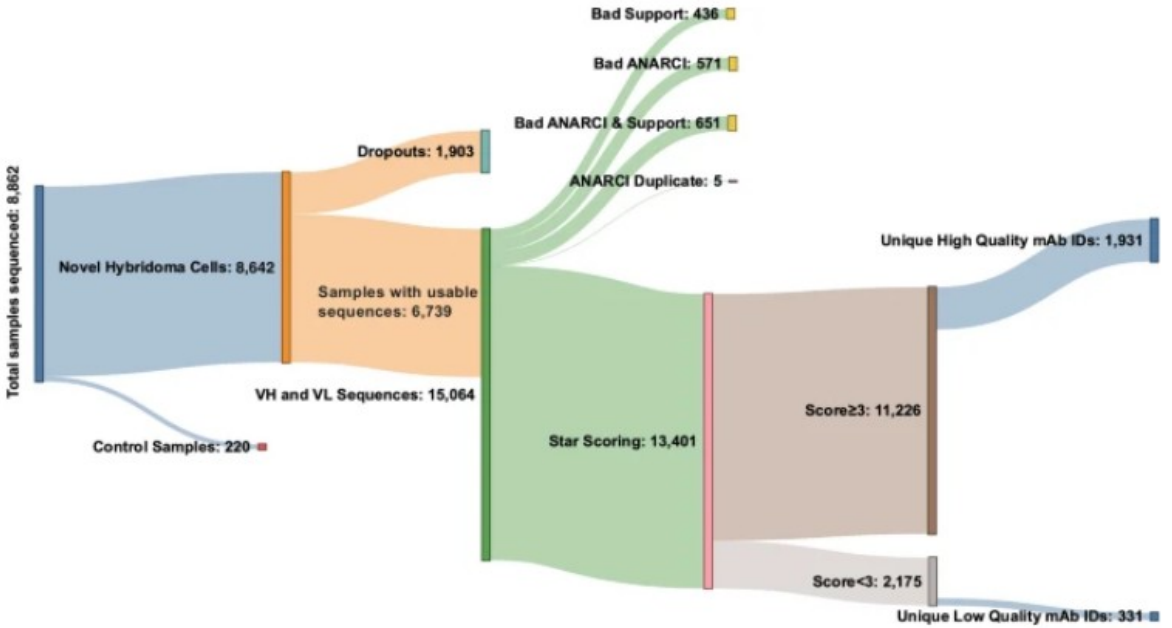
So, how do we get the sequences ?

From existing hybridomas. Originally hard to do, now can be done in high-throughput. Drawbacks : doesn't always express well / can change specificity.

High-volume hybridoma sequencing on the NeuroMabSeq platform enables efficient generation of recombinant monoclonal antibodies and scFvs for neuroscience research

Keith G. Mitchell, Belvin Gong, Samuel S. Hunter, Diana Burkart-Waco, Clara E. Gavira-O'Neill, Kayla M. Templeton, Madeline E. Goethel, Malgorzata Bzymek, Leah M. MacNiven, Karl D. Murray, Matthew L. Settles, Lutz Froenicke & James S. Trimmer

Scientific Reports **13**, Article number: 16200 (2023) | Cite this article



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De novo : multiple approaches, not mutually exclusive

Immunized humans
Animal models
Display technologies
In-silico design

Cell-sorting post immunization

Mining the human B cell repertoire: Flow cytometry + Single-cell sequencing.

Potently neutralizing and protective human antibodies against SARS-CoV-2

Seth J. Zost, Pavlo Gilchuk, James Brett Case, Elad Binshtein, Rita E. Chen, Joseph P. Nkolola, Alexandra Schäfer, Joseph X. Reidy, Andrew Trivette, Rachel S. Nargi, Rachel E. Sutton, Naveenchandra Suryadevara, David R. Martinez, Lauren E. Williamson, Elaine C. Chen, Taylor Jones, Samuel Day, Luke Myers, Ahmed O. Hassan, Natasha M. Kafai, Emma S. Winkler, Julie M. Fox, Swathi Shrihari, Benjamin K. Mueller, ... James E. Crowe Jr

+ Show authors

Nature **584**, 443–449 (2020) | Cite this article

Cell-sorting post-immunization/infection

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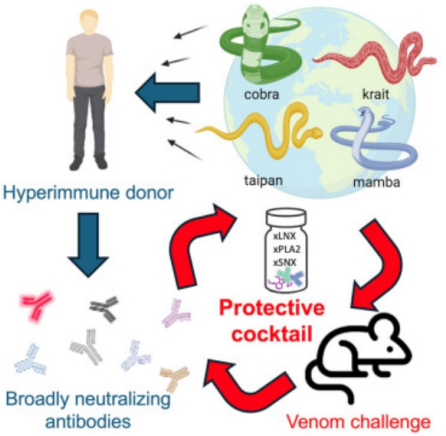
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Snake venom protection by a cocktail of varespladib and broadly neutralizing human antibodies

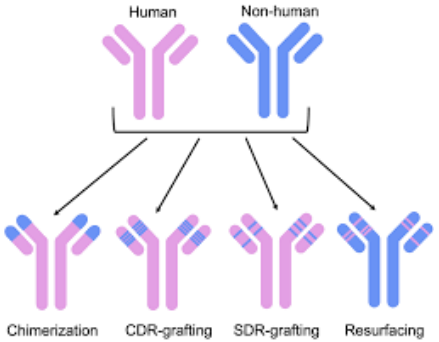
Jacob Glanville [1,4,5] · Mark Bellin [1,4] · Sergei Pletnev [2,4] · ... · Hannah Hirou [1] · Tracy Liu [2] · Peter D. Kwong [2,3] ... Show more



Immunization in animals

Prospects for the computational humanization of antibodies and nanobodies

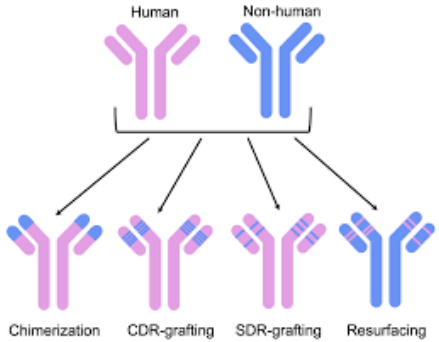
Gemma L. Gordon, Matthew I. J. Raybould, Ashley Wong and Charlotte M. Deane*



Chimerization (human constant regions) then CDR grafting (human frameworks, keeping only the six loops)

Fully human antibodies still elicit anti-idiotypic responses

Immunization in animals



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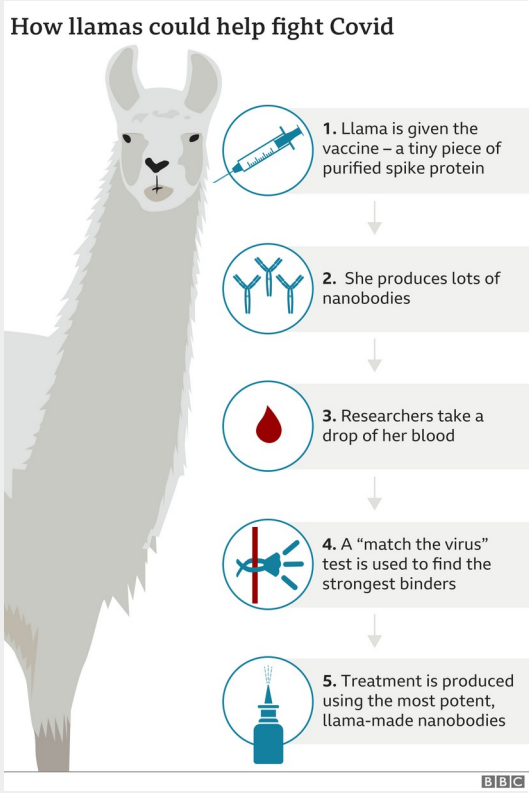
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Llamas and single-domain antibodies

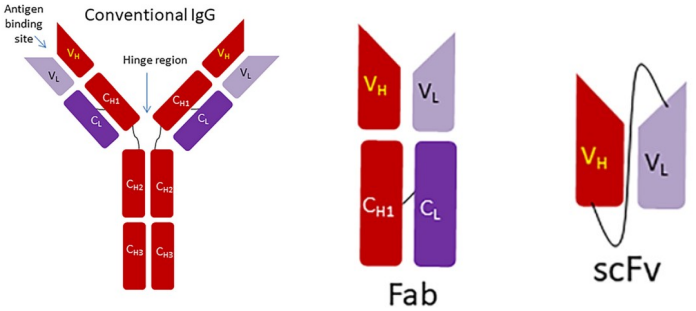
Nanobodies are easy to produce and test (one chain) and diffuse more than full Ab.

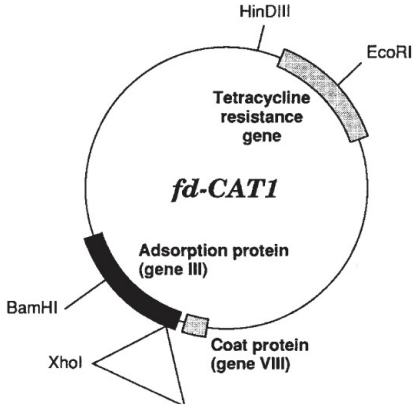


Display technologies

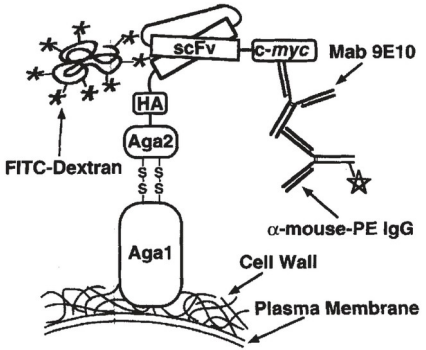
Full antibodies currently cannot be displayed directly on microbes or viruses.

Alternative formats (scFv, Fabs)





McCafferty et al., Nature 1990

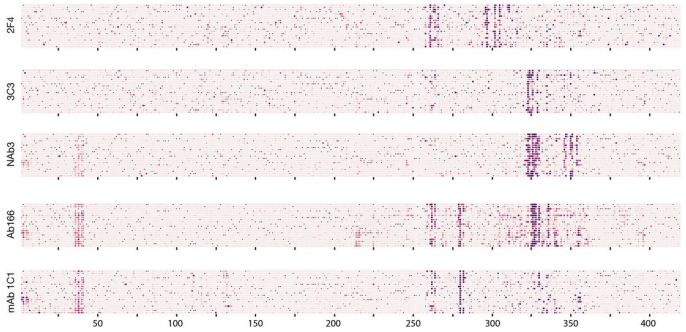


Boder, Wittrup, Nature Biotechnologies 1997

Phage & Yeast display methods can test up to 10^{11} “antibodies” at the same time.

Display constraints

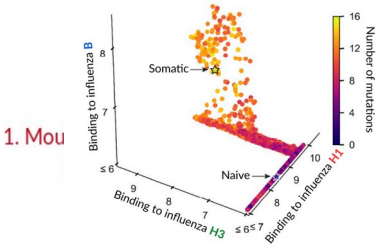
Display technologies are driven by the relatively low cost of DNA synthesis ⇒ we can design fully synthetic libraries.



Deep mutational scanning

Combinatorial libraries

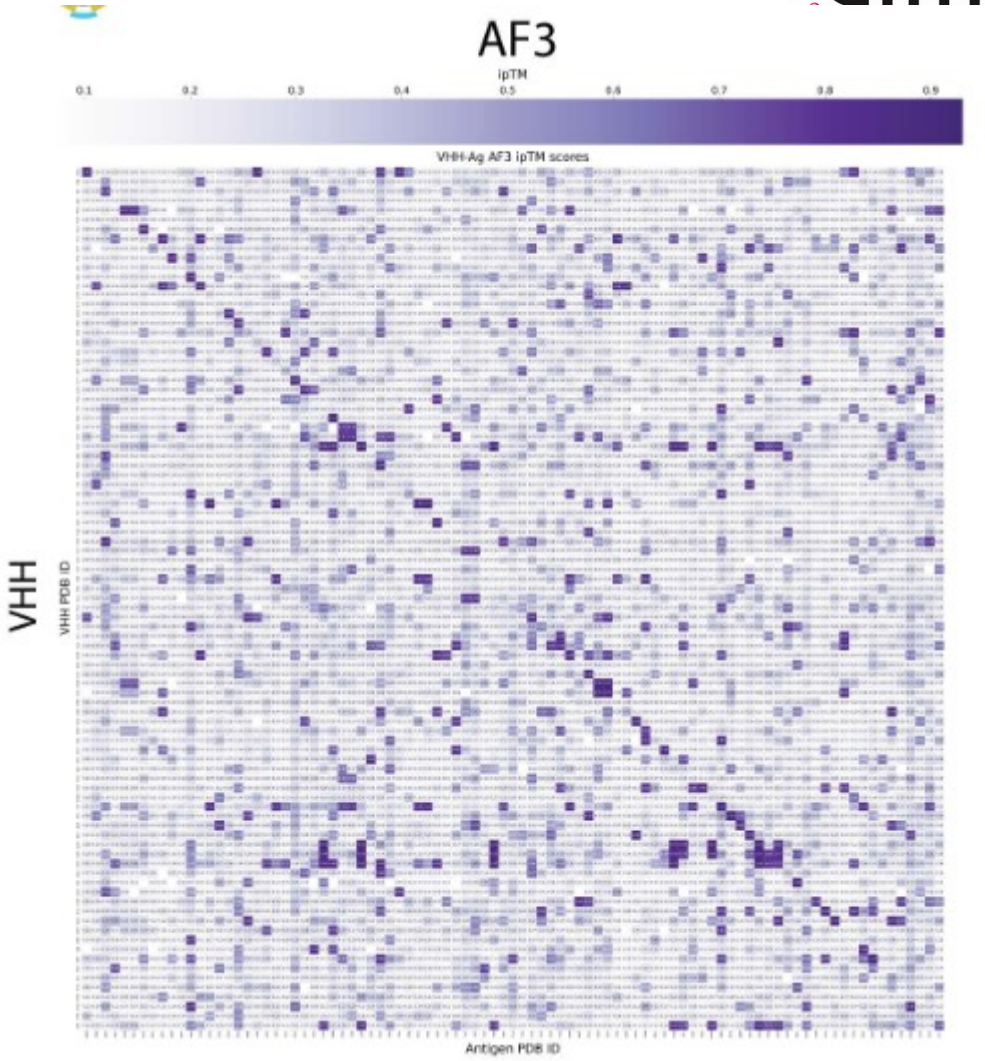




Limits: Folding differences, glycosylation, “**you get what you select for**”

In-silico design

Antibodies are particularly hard problems to solve numerically, even for the new structural prediction algorithms.



Structural Plausibility Without Binding Specificity: Limits of AI-Based Antibody-Antigen Structure Prediction Confidence Scores

Eva Smorodina [1,#,☯] (0000-0002-5457-5163), Montader Ali [2,☯] (0009-0004-9022-3896), Klara Kropivšek Brumat [3,☯] (0000-0003-1866-4094), Leonardo Salicari [4] (0000-0003-4444-3885), Samo Miklavc [5] (0009-0002-6381-8443), Aibek Kappassov [1] (0009-0004-1415-0082), Chengcheng Fu [6] (0009-0005-5551-6957), Pietro Sormanni [2,7] (0000-0002-6228-2221), Ario de Marco [3] (0000-0001-7729-819X), Victor Greiff [1,8,#] (0000-0003-2622-5032)

In-silico design

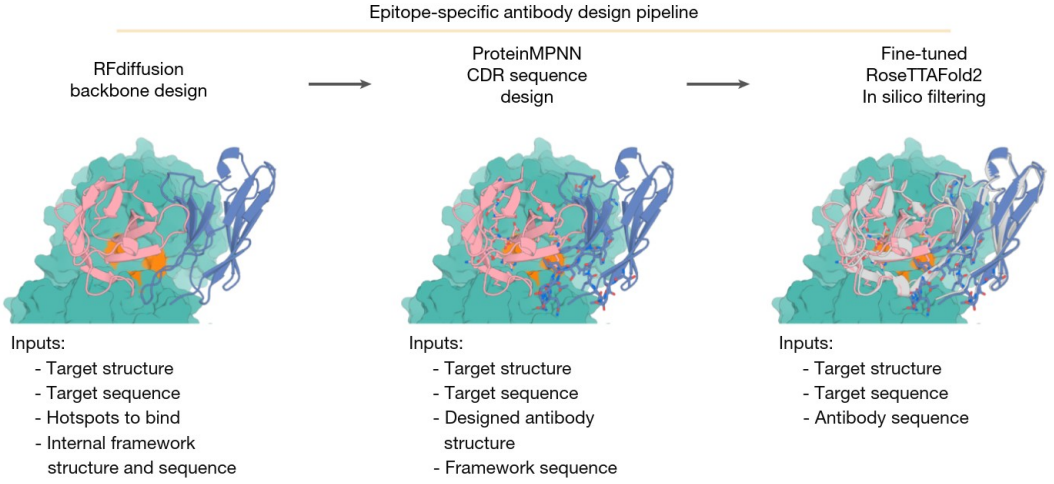
Binders against four disease-relevant epitopes, confirmed by cryo-EM.

9,000 designs tested per target (with display) and the generated antibodies had weak affinity (increased by in-vitro maturation)

Atomically accurate de novo design of antibodies with RFdiffusion

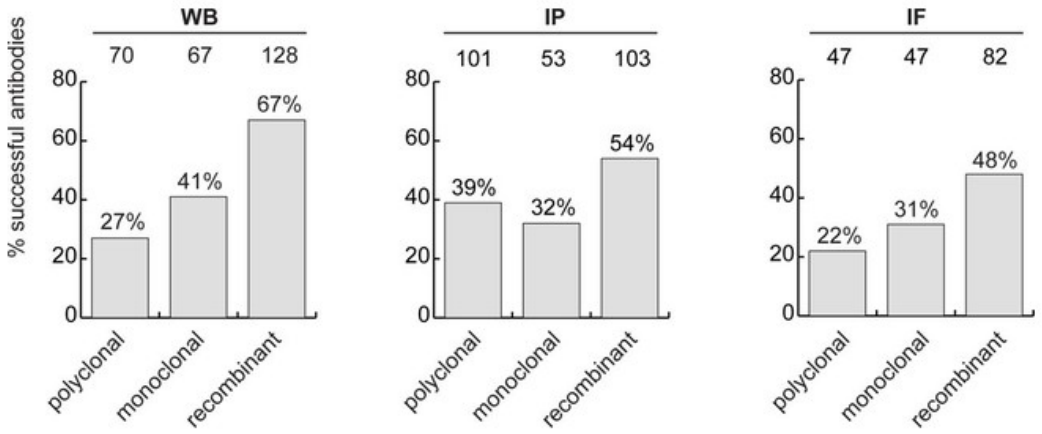
Nathaniel R. Bennett, Joseph L. Watson, Robert J. Ragotte, Andrew J. Borst, DéJenaé L. See, Connor Weidle, Riti Biswas, Yutong Yu, Ellen L. Shrock, Russell Ault, Philip J. Y. Leung, Buwei Huang, Inna Goreshnik, John Tam, Kenneth D. Carr, Benedikt Singer, Cameron Criswell, Basile I. M. Wicky, Dionne Vafeados, Mariana Garcia Sanchez, Ho Min Kim, Susana Vázquez Torres, Sidney Chan, Shirley M. Sun, ... David Baker + Show authors

Nature **649**, 183–193 (2026) | Cite this article

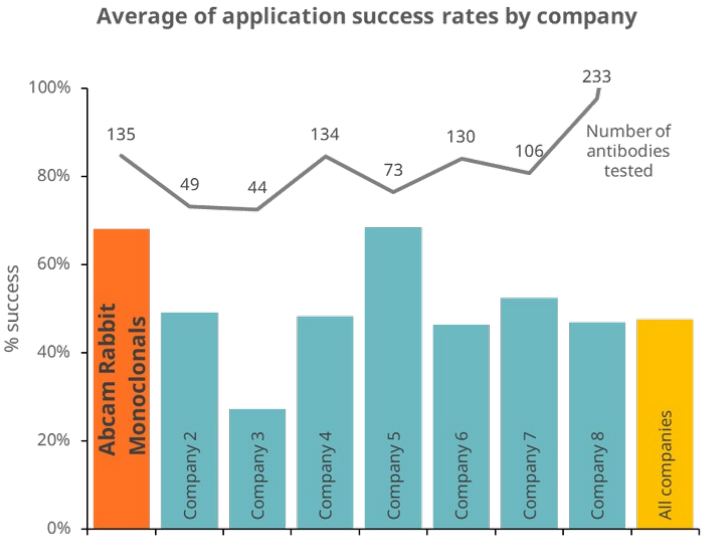


Reproducibility

Commercial antibodies only work 50% of the time.



YCHAROS
ANTIBODY CHARACTERIZATION THROUGH OPEN SCIENCE



Legalese

Article 4

Principle of replacement, reduction and refinement

1. Member States shall ensure that, wherever possible, a scientifically satisfactory method or testing strategy, not entailing the use of live animals, shall be used instead of a procedure.

At the European level, a recommendation to limit animal use was written in 2020 (proposed to ban ascites and no polyclonals if an alternative exists).



Non-animal-derived monoclonal antibodies are not ready to substitute current hybridoma technology

África González-Fernández, Francisco J. Bermúdez Silva, Marcos López-Hoyos, César Cobaleda, Lluís Montoliu, Margarita Del Val & Kirk Leech

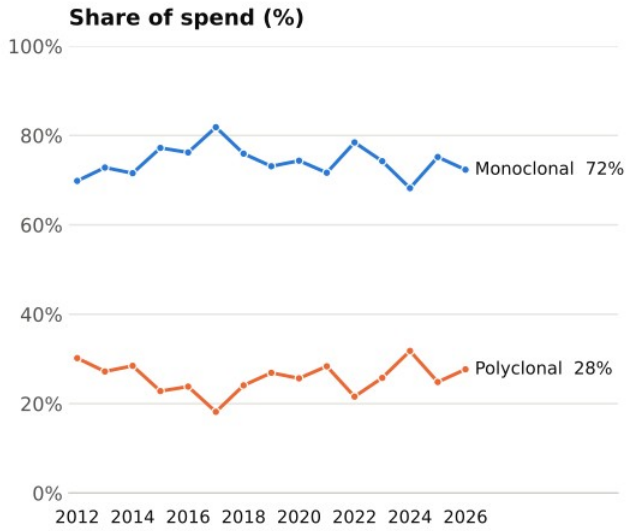
Nature Methods **17**, 1069–1070 (2020) | Cite this article

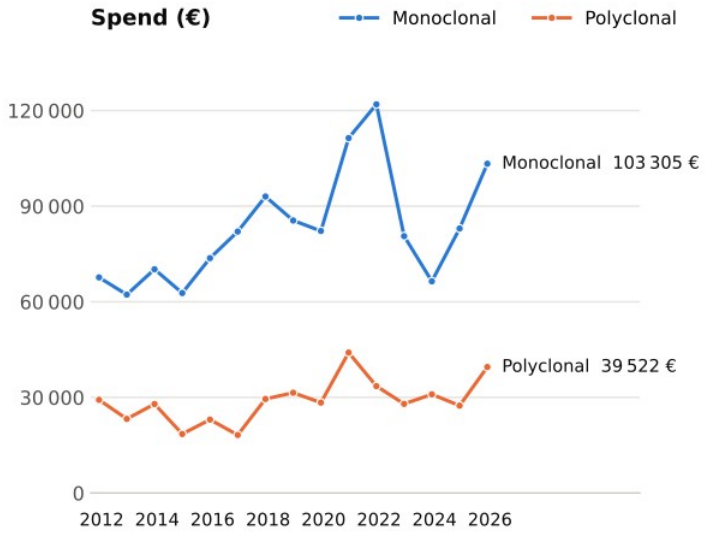
Industry pushback, from EARA (lobbying for animal research) and EFPIA (pharma lobbying) as well as some scientific pushback.

In 2023 the national committee subgroup decided this was premature and that there was an urgent need to do nothing.

What about CIML ?

Data extracted from Geslab. Approx 300k€ / year. This shows only antibodies identified as poly or mono (lots of unknowns).





For contrast, the split between mono-poly from (worldwide) publication data is 70-30%.

Summary

Three main types of production, relatively equal in volume: polyclonal, hybridomas and recombinants.

Recombinant antibodies are more reproducible, more engineerable, more patentable and have been completely adopted for drugs. For research reagents, change is slow due to inertia and the lack of economic incentive.